

Deepak Renugopal

Project Manager | AI Engineer | AI Solutions Specialist

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PROFESSIONAL SUMMARY

Project Manager with 9 years of enterprise IT delivery experience, transitioning into AI Engineering with hands-on expertise in machine learning, deep learning, NLP, computer vision, and AI deployment. Holds the Microsoft Azure AI Fundamentals (AI-900) certification and a Wharton Online Certificate in AI for Business. Built and deployed two production AI applications: an AI Radiology Report Summariser (NLP, fine-tuned T5) and a Visual Product Recommendation engine (computer vision, MobileNetV2) serving 31,000+ images. Proven ability to manage large-scale transformation programmes (Marks & Spencer, Legal & General) while applying AI tools for project tracking, predictive analytics, and reporting automation. PRINCE2 certified. Combines strong stakeholder management and delivery governance with practical AI engineering skills to drive data-driven, innovative solutions.

CORE COMPETENCIES

Project Management | AI Engineering | Machine Learning & NLP | Model Deployment | Workflow Automation | Predictive Analytics | Reporting Automation | Risk Management | Stakeholder Engagement | Team Leadership | Change Management | KPI & SLA Tracking | Business Process Mapping | Cloud Migration (Azure) | Agile / Scrum | PRINCE2 | Business Analysis | Strategic Planning

TECHNICAL SKILLS

Languages: Python, SQL, R

Web: HTML, CSS, JavaScript

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, Transformers (T5), MobileNetV2 / CNN Embeddings, Model Fine-Tuning, Cosine Similarity, ROUGE Evaluation

Cloud & Tools: Microsoft Azure, Render, HuggingFace Hub, Flask, FastAPI, REST API Development, Git/GitHub, MLOps

Project Tools: PRINCE2, Agile/Scrum, MS Project, JIRA

AI PROJECTS

AI Radiology Report Summariser | Personal Project

2026

- Designed and built an end-to-end NLP pipeline that ingests raw radiology reports and generates concise, plain-English summaries for clinical staff and patients.
- Fine-tuned a T5-base transformer summarisation model on 2,700+ chest X-ray reports, achieving strong evaluation scores (ROUGE-1: 0.55, ROUGE-L: 0.54).
- Deployed the solution as a FastAPI REST API on Render with the fine-tuned model hosted on HuggingFace, integrating a lightweight web front-end for real-time report upload and summary generation.
- Applied prompt-engineering techniques and post-processing rules to preserve critical clinical findings while removing technical jargon.
- Evaluated model performance using ROUGE metrics and iterated based on clinician feedback, demonstrating a complete MLOps lifecycle.
- Source code and documentation available at github.com/deepakrenugopal/medical-summarizer-api.

Visual Product Recommendation Engine | Personal Project

2026

- Built an end-to-end computer vision recommendation system that returns visually similar products from a catalogue of 31,000+ images, deployed as a live Flask web application.
- Generated deep feature embeddings using a pre-trained MobileNetV2 CNN and implemented cosine-similarity search to rank and retrieve the most visually similar items in real time.
- Engineered a scalable image-hosting architecture on HuggingFace, batching 31,000 images across structured subfolders with a JSON filename index to overcome platform file limits.
- Hosted the MobileNetV2 model on a HuggingFace Space and called it via API from the Flask backend on Render, keeping the web application within free-tier memory limits.

- Deployed to Render with memory-efficient, float16 embedding storage, demonstrating practical trade-off management between model performance and cloud resource constraints.
- Source code and documentation available at github.com/deepakrenugopal/visual-similarity-recommender.

PROFESSIONAL EXPERIENCE

IT Support (Backfill) | Roche Diagnostics, Burgess Hill

March 2026 – Present

- Delivering on-site IT support for approximately 1,000 users within a regulated healthcare diagnostics environment, managing and resolving incidents and service requests via ServiceNow in line with SLA targets.
- Imaging, configuring, and deploying laptops to end users, maintaining standardised builds and hardware asset records to ensure consistent device performance.
- Administering Active Directory including user account creation, password resets, group policy management, and access provisioning for on-site staff.
- Supporting printer fleet management and mobile device configuration, ensuring seamless connectivity and device availability across the site.
- Serving as primary on-site IT resource within a 5-member team during scheduled absences, maintaining uninterrupted service continuity and end-user satisfaction in a fast-paced regulated environment.

AI Upskilling & Project Development | Self-Directed Learning

Jan 2024 – Present

- Completed AI Full Stack Training covering machine learning, deep learning, NLP, model deployment, and AI-assisted business workflows.
- Attained Microsoft Azure AI Fundamentals (AI-900) certification.
- Built and deployed two production AI applications: the AI Radiology Report Summariser (NLP) and a Visual Product Recommendation engine (computer vision).
- Completed Wharton Online Certificate in Artificial Intelligence for Business (University of Pennsylvania, 2025).

Project Manager | Tata Consultancy Services

Sept 2015 – Jan 2024

Marks & Spencer

- Managed end-to-end Windows 10 migration for approximately 3,000 end users, including project planning, scheduling, governance, and stakeholder communication.
- Served as Incident Manager for the .com transformation project, coordinating cross-functional onshore and offshore teams to migrate consumers from in-store to online platforms.
- Developed and maintained detailed project documentation, progress reports, and communication plans to ensure transparency and issue escalation management.
- Facilitated regular project ceremonies and decision-making forums to maintain momentum and alignment with business goals.

Legal & General

- Oversaw Windows 10 migration and lifecycle management for approximately 8,000 end users across the UK, Ireland, and Bermuda, managing roadmaps, governance, financials, resource allocation, and change management.
- Served as Incident Manager for the Group Finance business unit, leading resolution of low-to-high priority tickets and managing a team of 10 service technicians.
- Facilitated Agile ceremonies including sprint planning, daily scrums, backlog refinement, and sprint reviews across multi-year transformation programmes.
- Led cloud migration transitioning approximately 50 servers to Microsoft Azure, coordinating technical and business stakeholders throughout.
- Managed vendor relationships and application lifecycle activities including renewals and technology ecosystem integration.
- Developed comprehensive project documentation, schedules, and stakeholder communications to align cross-functional teams and senior management.

IT Support Engineer | HCL Technologies

Aug 2010 – Aug 2015

- Led data centre process transition to a new facility in Singapore, driving operational continuity and efficiency improvements.
- Designed and delivered comprehensive IT and data centre training programmes across multiple global locations.
- Collaborated with on-site engineers to resolve operational issues and major incidents, ensuring minimal service disruption.

- Acted as primary customer liaison, gathering feedback to identify service enhancement opportunities.
- Mentored team members in bug resolution, process enhancements, and best-practice adoption.

Helpdesk Support | Dell Technologies

2005 – 2010

- Provided first-line technical support and issue resolution, building a strong foundation in IT systems and customer service delivery.

EDUCATION & CERTIFICATIONS

Bachelor of Engineering, Computer Science

2005

Anna University, India

Microsoft Azure AI Fundamentals (AI-900)

2026

Microsoft

Certificate, Artificial Intelligence for Business

2025

Wharton Online / University of Pennsylvania

PRINCE2 Project Management, Foundation & Practitioner

Certified

AXELOS